

GSP TECHNICAL SPECIFICATION SHEET

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GLASS SUBSTRATE PROTOCOL™ (GSP)

Developer-Facing Architecture Specification

1. Purpose

GSP defines the structural behavior of AI and digital systems through:

- standard event logging
- consistent state transitions
- digital substrate formation
- cryptographic architecture-level verification

2. Core Structural Components¹

2.1. Event Substrate Layer (ESL)

Defines how system events are encoded, formatted, sequenced, and stored.

2.2. State Consistency Engine (SCE)

Ensures deterministic state transitions and prevents drift.

2.3. Cryptographic Attestation Grid (CAG)

Provides:

- hash-linked events
- tamper-evident structure
- verifiable system pathways

2.4. Reconstruction Protocol (RPX)

Allows full sequence reconstruction from substrate logs.

¹ November 2025

3. Technical Behaviors

- Event integrity verification
- Substrate-level timestamps
- Hash-chain sequencing
- Backward-compatible structure maps
- Standardized metadata schema
- Cross-platform substrate uniformity

4. Compliance Requirements

To be **Sierra Certified™**, a GSP implementation must provide:

- Full ESL coverage
- CAG-compliant cryptographic proof
- SCE-consistent state transitions
- RPX reconstructability
- Compatibility with G-BALO audit tools

5. API Architecture

Endpoints Include:

- `/substrate/event` — event submission
- `/substrate/log` — retrieve structural log
- `/substrate/hash` — verify hash integrity
- `/substrate/state` — fetch current state profile
- `/substrate/reconstruct` — perform RPX reconstruction

6. Deployment Models

- On-prem enterprise
- Cloud structured
- Hybrid GSP-compliant systems
- Multi-agent architecture integration
- Judicial evidence pipeline integration